Day 11: VLSM Review and the OSI Model

Level 1, Month 1, Networking foundations

What I learned

The OSI model splits the work of a network into 7 layers, and each layer has one job. Counting from the bottom, layer 1 is the closest to the cable.

Layer	Name	Role	Example
7	Application	Services used by programs	HTTP, DNS
6	Presentation	Data format and encryption	TLS
5	Session	Opening and closing sessions	
4	Transport	End-to-end delivery, ports	TCP, UDP, port 443
3	Network	Logical addressing and routing	IP address, router
2	Data Link	Local delivery	MAC address, switch
1	Physical	Signal on the medium	Cable, RJ45

I can already see these layers in Wireshark. A ping packet shows `Ethernet II` (layer 2, with the MAC), `IPv4` (layer 3, with the IP and TTL), and `ICMP` on top of layer 3.

Review problem: VLSM on 172.16.10.0/24

Hosts	Bits	Mask	Block	Network	First	Broadcast	Last
120	7	/25	128	172.16.10.0	.1	.127	.126

Hosts	Bits	Mask	Block	Network	First	Broadcast	Last
60	6	/26	64	172.16.10.128	.129	.191	.190
25	5	/27	32	172.16.10.192	.193	.223	.222
2	2	/30	4	172.16.10.224	.225	.227	.226

- Used: 128 + 64 + 32 + 4 = 228 addresses.
- Free: 256 - 228 = 28 addresses (172.16.10.228 to 172.16.10.255).
- Everything was correct, with no mistakes.

OSI exercise

Item	Layer
MAC address	2
IP address	3
RJ45 cable	1
Port 443 and TCP	4
Router	3

Error log

No technical mistakes today. One thing to improve: my answers to the OSI exercise were written as `1-2`, `3-2`, and so on, and the format was ambiguous. Next time I write it as `item -> layer`, for example `MAC -> 2`.

Main lessons

- Clear handwriting and one step per line removed the reading errors from before.
- Layers help me place every protocol: where MAC lives, where IP lives, and where TCP lives.

- The 28 free addresses are the gain of VLSM over using one mask for everything.

Anki

New cards in `anki-cards-11.txt`.

Next session

The TCP/IP model and encapsulation: how a message becomes a packet and a frame, read in Wireshark.